2024—2025春高数下期末答案(回忆版)

1. (1) D; (2) C; (3) A; (4) D; (5) 图A.

2. (1) $2x-y-z=1$; (2) $\frac{\pi}{2}$; (3) $\sqrt{5}$; (4) $\frac{1}{2}$; (5) 2.

3. (1) $\frac{dy}{dx}=\frac{\sin t}{1-\cos t}$

When $t=\frac{\pi}{3}$, the corresponding point is $(\frac{\pi}{3}-\frac{\sqrt{3}}{2},\frac{1}{2})$.

$$\sqrt{3}x-y=\frac{\pi}{\sqrt{3}}-2.$$

(2) $\frac{d^2y}{dx^2}=-\frac{1}{(1-\cos t)^2}$.

4. (1) Applying the root test, we obtain that the series converges obsolutely.

(2) The alternating series test gives that the series converges, while the integral test leads to the conclusion that the series does not converge absolutely. Therefore, the series converges conditionally.

5. We first identify critical points by solving the gradient equations:

$$\begin{cases}\frac{\partial f}{\partial x}=(1-x^2)e^{-\frac{x^2+y^2}{2}}=0\\ \frac{\partial f}{\partial y}=-xye^{-\frac{x^2+y^2}{2}}=0\end{cases}$$

Solving these equations yields two critical points $(1,0)$ and $(-1,0)$.

Next, we employ the second derivative test. Computing the Hessian component

$$f_{xx}(x,y)=(x^3-3x)e^{-\frac{x^2+y^2}{2}}$$
$$f_{xy}(x,y)=(x^2-1)ye^{-\frac{x^2+y^2}{2}}$$
$$f_{yy}(x,y)=x(y^2-1)e^{-\frac{x^2+y^2}{2}}$$

For the critical point $(1,0)$,

$$f_{xx}(1,0)=-2e^{-\frac{1}{2}}<0,\ f_{xy}(1,0)=0,\ f_{yy}(1,0)=-e^{-\frac{1}{2}}.$$

Then, $\Delta=f_{xx}f_{yy}-f_{xy}^2=(-2e^{-\frac{1}{2}})(-e^{-\frac{1}{2}})-0=2e^{-1}>0.$

Therefore, $(1,0)$ is a local maximum point with $f(1,0)=e^{-\frac{1}{2}}$.

Similarly, at $(-1,0)$,

$$f_{xx}(-1,0)=2e^{-\frac{1}{2}}>0,\ f_{xy}(-1,0)=0,\ f_{yy}(-1,0)=e^{-\frac{1}{2}}.$$
$$\Delta=(2e^{-\frac{1}{2}})(e^{-\frac{1}{2}})-0=2e^{-1}>0$$

Thus, $(-1,0)$ is a local minimum point with $f(-1,0)=-e^{-\frac{1}{2}}$.

In summary, the function attains its local maximum value $e^{-\frac{1}{2}}$ at $(1,0)$, and the local minimum value $-e^{-\frac{1}{2}}$ at $(-1,0)$.

6. From the symmetry, we obtain that $\bar{x}=\bar{y}=0$.

$$M_{xy}=\iiint_D z\,dxdydz=\int_0^{2\pi}\int_0^1\int_{r^2}^1 zr\,d\theta dr dz=\int_0^{2\pi}d\theta\int_0^1 r(\frac{1}{2}-\frac{r^4}{2})dr=\frac{\pi}{3}.$$
$$M=\iiint_D dxdydz=\int_0^{2\pi}\int_0^1\int_{r^2}^1 r\,d\theta dr dz=\int_0^{2\pi}d\theta\int_0^1 r(1-r^2)dr=\frac{\pi}{2}.$$
$$\bar{z}=\frac{M_{xy}}{M}=\frac{2}{3}$$

Therefore, the centroid is $(0,0,\frac{2}{3})$.

7. $$\nabla\cdot F = 6x^2+3y^2$$

Thus the flux is given by

$$\iiint_0 \nabla\cdot F\,dV = \iint_{0\le 2x^2+y^2\le 1} (1-(2x^2+y^2))(6x^2+3y^2)\,dx\,dy.$$

Apply the transformation $x=\frac{1}{\sqrt{2}}u$, $y=v$, we obtain

$$= \frac{3}{\sqrt{2}} \iint_{0\le u^2+v^2\le 1} (1-(u^2+v^2))(u^2+v^2)\,du\,dv$$

$$= \frac{3}{\sqrt{2}} \int_0^{2\pi}\int_0^1 (1-r^2)r^3\,dr\,d\theta$$

$$= \frac{\sqrt{2}}{4}\pi.$$

8. $$\nabla\times F = \langle 0,-1,2\rangle.$$

Let S be the region $\{(x,y,z)\,|\,x^2+y^2\le 2y,\ x-y+z=2\}$. Obviously C is the boundary of S with the corresponding unit normal vector $n=\frac{1}{\sqrt{3}}\langle 1,-1,1\rangle$. For the surface S, we have

$$d\sigma = \sqrt{3}\,dx\,dy$$

Applying Stokes' Theorem, the equivalent surface integral is obtained as

$$\iint_S \nabla\times F\cdot n\,d\sigma = \iint_{x^2+y^2\le 1} 3\,dx\,dy = 3\pi$$